

1. Supply 26

1. Apply union-by-rank & path compression techniques by UNION & FIND operations.

2. Analyze quicksort using divide & conquer and derive its best, worst & average time complexities.

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3. (i) Apply mergesort to sort the list :

$a[1:10] = (31, 28, 17, 65, 35, 42, 86, 25, 45, 52)$

Draw the tree of recursive calls of merge sort & merge functions.

(ii) Experiment with collapsing find and splitting find to determine which produces better results when used in conjunction with weighted union.

4. i) illustrate the best, average, worst case time complexity of quicksort with example
- ii) demonstrate the key steps involved in applying divide & conquer & analyze its time complexity.

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5. Explain disjoint set operations with UNION-FIND Algorithms.

6. write an algorithm for quicksort using divide & conquer and illustrate the process with an example.

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7. construct disjoint sets for a sequence of graph operations and show how UNION-FIND helps detect cycles.

8. Analyze strassen's matrix multiplication using divide & conquer & derive its recurrence relation

9. Explain the role of UNION & FIND ~~algorithms~~ operations in the disjoint set data structure.

10. perform binary search on list of elements to find the key element using divide & conquer, and also estimate the time complexity.

i) What is weighting rule for union, how it improves the performance of union operation.

ii) describe iterative binary search algorithm with an example.

i) Explain the Strassen's matrix multiplication approach & derive its time complexity.

ii) find & explain the time complexity of quicksort in both cases.

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13. Explain connected & biconnected components in the disjoint set data structure.
14. write divide & conquer recursive merge sort algorithm & define the time complexity of algorithm

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15. Elaborate about disjoint set operations with an example.
16. Explain the working of binary search algorithm using divide & conquer with an example.
17. Elaborate about connected components & biconnected components of a graph.
18. distinguish b/w quicksort & merge sort & arrange the following numbers in increasing order using quicksort
- 98, 78, 34, 12, 64, 10